Lab report

Audio Amplifier

EE23BTECH11027 [1] **and EE23BTECH11042** [2]

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Abstract

In this experiment, we have used LM386 IC. It has an output power of up to $1W$ and is therefore very handy in portable audio applications for small speakers. Its gain can be configured anywhere between 20 and 200, making it suitable for connecting to virtually any input source, while maintaining low distortion and good sound quality. It is just the right price and easy enough to build, perfect for classes and other DIY projects.

1. Introduction

This report details the design,working,and analysis of a simple audio amplifier circuit using the LM386 IC. The circuit amplifies weak audio signals from a condenser microphone and outputs an amplified signal through a speaker.

2. Materials Required

Component	Specification	Quantity
LM386	Audio Amplifier IC	1
Microphone	-	1
Speaker	8Ω, 0.5W	1
Resistors	10kΩ, 100kΩ	2
Capacitors	0.1μF, 10μF, 0.05μF, 220μF	5
Potentiometer	100kΩ	1
Power Supply	5V DC	1

Table 1. List of materials required for the audio amplifier circuit.

3. Circuit Diagram

The circuit diagram for the audio amplifier is shown below:

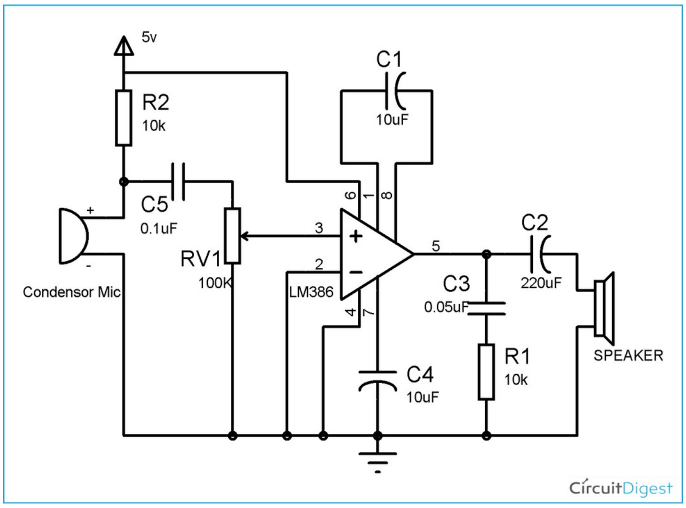

Figure 1. Audio Amplifier Circuit Diagram.

4. Working

1. **Microphone Input**: The condenser microphone captures audio signals and converts them into equivalent electric signals.

2. **Pre-amplification**: Resistor R_2 and capacitor C_5 offer biasing and filtering of the microphone signal. The signal then goes into the inverting input of the LM386 through the potentiometer R_{V1}.

3. **Amplification**: The LM386 amplifies the input signal. The gain is controlled by the capacitor C_1 connected between pins 1 and 8. The gain is maximized to 200 on connecting them with $10\mu F$ capacitor.

4. **Output Stage**: The amplified signal is passed through capacitor C_2 for DC blocking and fed to the speaker for audio output.

5. Calculations

Gain: The LM386 amplifier has a voltage gain given by:

$$\text{Gain (dB)} = 20\log\left(\frac{V_{\text{out}}}{V_{\text{in}}}\right)$$

Bandwidth: The gain deviation is within -3 dB across the range of 20 Hz to 20 kHz, indicating the circuit's effective audio bandwidth.

Noise: The total noise voltage is calculated as the RMS value of noise across the frequency range.

6. Frequency Response and Noise Analysis

The frequency response and noise characteristics of the circuit were analyzed, and the results are shown below:

35 **Gain vs. Frequency**

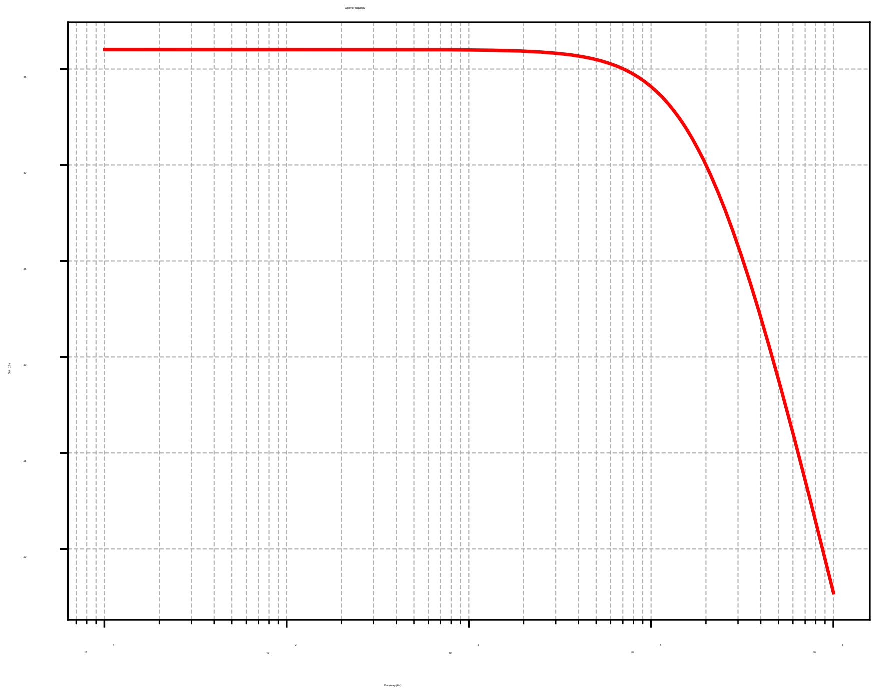

Figure 2. Gain vs. Frequency plot.

36 **Noise vs. Frequency**

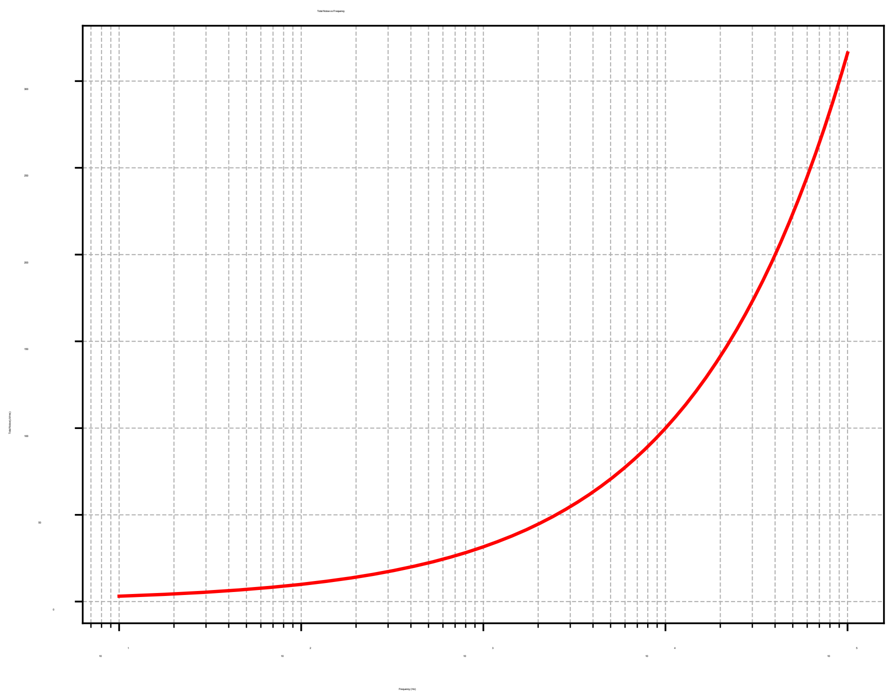

Figure 3. Noise vs. Frequency plot.